

CANDIDATE TAKE-HOME ASSIGNMENT

Applied AI Engineer — Task Sheet

Generative AI / LLM · Make My Education

Estimated effort: To be submitted by 15th July, 2026, Wednesday, 11 AM.

1 About Make My Education

This document is your take-home assignment for the Applied AI Engineer (Generative AI / LLM) role. It also gives you the context you need about who we are and what we're building, so your submission can speak directly to our problem.

Who we are

Make My Education is an Indian EdTech platform that helps students discover and compare colleges and courses, filter by course and location, browse detailed college profiles, connect with admission counsellors, and apply — all in one place, on web and on our Android app.

Where we're going

We're evolving from a search-and-listing product into an AI-native admissions platform: a personal AI counsellor for every student, grounded in our verified college data, available in their own language, that not only advises but helps them shortlist and apply. This role owns that AI — and your take-home is a small, honest slice of that exact problem.

Company snapshot

Field	Detail
Company	Make My Education India Private Limited
Platform	MakeMyEducation — college discovery, comparison, counselling & admissions (web + Android app)
Sector	EdTech / higher-education admissions (India)
Corporate office	Unit No. 2, Ground Floor, H-201, Sector-63, Noida – 201309, Uttar Pradesh
Website	www.makemyeducation.com
Phone	+91 73029 83393
Email	contact@makemyeducation.com
CIN	U85500UT2026PTC021127

2 What We're Looking For

We're hiring a builder, not a researcher. We care far more about whether you can ship a reliable, cost-aware AI feature that real students would trust than about publications or benchmark scores. Specifically:

- **Product sense:** you think about the student's real problem, not just the model.
- **Cost-pragmatism:** you make sensible trade-offs on tokens, models, and infrastructure — and can explain them with measured numbers.
- **A grounding mindset:** you know that in admissions a wrong fee or cutoff destroys trust; accuracy beats fluency.
- **Shipping ability:** you've put AI in front of real users and dealt with what happens next.
- **Clear thinking:** simple, well-reasoned solutions over clever-but-fragile ones.

3 How We'll Evaluate

We're sharing our rubric up front so you know where to spend your time.

Criterion	What we're looking for	Weight
Grounding & accuracy	Answers come from the data and cite it; no invented facts; a graceful "I don't know".	25%
Retrieval & routing quality	Sensible chunking, embeddings and retrieval — and sound judgment about which questions need retrieval at all.	20%
Evaluation & testing	You measured your own system, found its failures, and told us about them.	15%
Cost & pragmatism	Reasoned model and infra choices, backed by measured numbers from your prototype.	15%
Code clarity & runnability	Readable, simple, runs from a clean checkout — with a clear README.	15%
Product thinking	Your reflection shows you understand our student and our business.	10%

We will run your code against a set of questions you have not seen. They test the same capabilities as the published questions — grounding, refusal, filtering, comparison — on different values. Build for the general case, not for the seven examples below.

4 Part A — Build a RAG Prototype (main task)

Your task

Build a small Retrieval-Augmented Generation (RAG) prototype that answers natural-language questions about colleges, using only the sample dataset we provide. Every answer must be grounded in the data and cite which record(s) it came from.

The dataset is attached as sample_colleges.csv — a synthetic set of 15 colleges. The full file is attached, together with DATA_DICTIONARY.md. Values are invented for this exercise; treat them as if they were real and verified.

Data dictionary — please read this before you start

Column	Type	Meaning
college_id	string	Stable identifier, C001–C015. Use this in your citations.
name	string	Full college name. Two entries have deliberately similar names.
city / state	string	Location.
type	enum	Government Private Deemed
courses_offered	string	Semicolon-separated. A Diploma is not a degree.
annual_fees_inr	integer	Tuition in ₹ per academic year. Not per semester. Excludes hostel, mess and any charges described in "about".
last_year_cutoff_pct	integer	Minimum aggregate percentage admitted last year. Treat as a hard floor — a student below this figure was not eligible.
total_seats	integer	Total intake across all courses.
hostel_available	boolean	Yes No.
naac_grade	string	A++, A+, A, B++, B+, B.
avg_placement_lpa	float	Average package in lakhs per annum. 0 means not reported / not applicable. 0 does NOT mean "worst placements".
established_year	integer	Year founded.
about	free text	~110 words per college: admission process, scholarships, hostel arrangements, extra charges, placement context, faculty. Unstructured. Some questions can only be answered from this field, and it sometimes qualifies or explains a structured value.

What to build

- Ingest the dataset and build a retrieval layer (embeddings + a vector store of your choice).
- Answer free-text questions by retrieving the relevant records and generating a grounded response with the foundation model of your choice.
- Cite the source college_id(s) for every factual claim in the answer.
- When the data can't answer a question, say so clearly — never invent a fact.
- Include a short README: how to run it, your key design choices, and — importantly — how you'd control cost at scale, with the measured numbers requested in Part D.

Required interface

So that we can run every submission the same way, please expose exactly this:

```
pip install -r requirements.txt
export OPENAI_API_KEY=... # or whichever provider you use — document it
python answer.py "Which colleges offer an MBA, and what do they cost?"
```

...printing a single JSON object to stdout:

```
{
  "answer": "Ganga Valley University (Rs 98,000/yr) and Doon Business School (Rs 1,75,000/yr).",
  "citations": ["C002", "C004"],
  "answered": true,
  "reason_if_unanswered": null
}
```

- Set `answered: false` when the data cannot support an answer; citations may then be empty.

Required deliverables in the repository

- **answers.md** — the verbatim output of your system for the seven published questions below. Copy-pasted, not edited.
- **run_all.py** — regenerates `answers.md` in one command.
- **evals/** — 8–10 question/expected-answer pairs of your own, plus a script that runs them and prints a pass rate. Tell us your score and where it fails. A candidate who ships a failing test they wrote themselves tells us more than one who ships a perfect demo.
- **README.md** — design choices, cost numbers (Part D), and what you would do differently with more time.

Questions your prototype should handle

- “I scored 78% and have a budget of ₹1.5 lakh/year — which engineering colleges can I consider?”
- “Which colleges offer an MBA, and what do they cost?”
- “List the government colleges that have hostel facilities.”
- “What’s the average placement package at North Ridge Institute of Technology?”
- “Does Ganga Valley University offer a PhD in Physics?” — (not in the data; it should not guess.)
- “Which colleges offer scholarships for students from low-income families?” — (answerable only from the free-text “about” field.)
- “Which college is best for me? I have ₹1 lakh per semester.” — (the unit differs from our data, and “best” is a judgment. Show us how you handle both.)

Scope — please keep it simple

- **Out of scope, and we award no marks for:** UI polish, authentication, deployment, fine-tuning, and multi-agent frameworks. A script, notebook, or minimal endpoint is fine.
- Use any language, model, and vector store you like; a hosted model API is perfectly acceptable.
- Note what you’d do differently with more time.

5 Part B — Proof You’ve Shipped

Share evidence of at least one LLM/AI feature you’ve put in front of real users. A link, demo, or short write-up is fine. Please tell us:

- What it did and who used it (a rough sense of scale helps).
- Your specific role — what you personally built.
- What broke or surprised you in production, and how you handled it.

- What it cost to run, and anything you did to bring that down.

6 Part C — Short Written Reflection

Keep this brief — a few sentences each. We care about your reasoning, not length.

- How would you keep per-query cost low for our AI counsellor as usage grows?
- How would you stop the system from ever stating a wrong fee or cutoff?
- If you joined tomorrow, what would you build first for Make My Education, and why?
- How would you measure whether AI is actually helping students?

7 Part D — Cost, With Numbers

Cost is 15% of your score, and we are asking for measurements, not intentions. Anyone can write "I would cache and use a smaller model." Very few instrument it. From your own prototype, please report:

Metric	Your number
Average input tokens per query	
Average output tokens per query	
Average end-to-end latency per query	
Model(s) used, and cost per 1M tokens	
Cost per 1,000 queries, in ₹	
One-time embedding cost for the full dataset	

Then, in three or four sentences: at 50,000 student queries a month, what breaks first — cost, latency, or accuracy? What would you change first, and roughly what would it save?

Rough measurement is fine. `time.perf_counter()` and the token counts your provider already returns in the API response are enough. We are looking for the instinct to measure, not for precision.

8 How to Submit

- Send a link to a Git repository (GitHub/GitLab) with your code and README.
- Include your Part B proof, Part C reflection and Part D cost table in the README or a short separate document.
- Required: A 2–3 minute screen recording walking through your prototype and showing it answer two of the published questions live.
- **Timeline:** Please submit it by Wednesday, 15th July, 2026 by 11 AM.
- **Next step:** A 30-minute call in which we run some questions you have not seen against your code, and ask you to make one small change live.
- **Questions:** For queries, you can email us at contact@makemyeducation.com.

We respect your time and read every submission ourselves. Thank you for your interest in building the future of college admissions with Make My Education.